

# National University of Computer and Emerging Sciences

## Lahore Campus

### Part (a) [10 Marks]

```
#include <iostream>
#include <cstring>
using namespace std;
class Person {
private:
    char* name;
public:
    Person(const char* n) {
        name = new char[strlen(n) + 1];
        strcpy(name, n);
        cout<<"Person() Called.\n";
    }
    ~Person() {
        if(name) delete[] name;
    }
    void displayName(){
        cout << "Name: " << name << endl;
    }
    Person(Person& rhs){
        cout<<"Copy Person Called.\n";
        name = rhs.name;
    }
    Person& operator=(Person& rhs){
        cout<<"Op = Called.\n";
        name = rhs.name;
    }
};

int main() {
    Person person1("Alice");
    Person person2 = person1;
    {
        Person person3 = person1;
        cout << "Person 3: ";
        person3.displayName();
        cout<<"-----\n"; //just
draw a small line
    }
    cout << "Person 1: ";
    person1.displayName();
    cout << "Person 2: ";
    person2.displayName();
    return 0;
}
```

### Part (b) [10 Marks]

```
#include <iostream>
using namespace std;
class Box {
    int* length;
public:
    Box(){
        length = new int;
        *length = 99;
    }
    Box(Box& rhs){
        length = new int;
        *length = *rhs.length;
    }
    Box& SomeFunction(int i){
        *length = i;
        return (*this);
    }
    ~Box(){
        cout << "Box destroyed. Length
was: " << *length << endl;
        if(length != 0)
            delete length;
    }
    void Print(){
        cout<<*length<<endl;
    }
    void AnotherFunction() {
        Box temp;
        *temp.length = 555;
        cout<<"Printing... "; Print();
        *this = temp;
        //temp deleted
    }
};

void main() {
    {
        Box obj1;
        {
            Box* obj2 =
            &(obj1.SomeFunction(10));
            cout<<"Obj 2 = ";
            obj2->Print();
        }
        cout<<"Obj 1 = ";
        obj1.Print();
    }
    cout<<"-----\n"; //just draw a
small line
    {
        Box obj1;
        obj1.AnotherFunction();
        cout<<"Obj 1 = ";
        obj1.Print();
    }
}
```

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### **CLO # 2: Apply OOP concepts (Encapsulation, Inheritance, Polymorphism, Abstraction) to computing problems for the related program**

#### **Q2: [20 marks]**

You are required to implement a class **MyList** that manages a dynamically growing array of strings. Partial class definition, driver program (main) and required output is given below. **Implement the class MyList such that the given main program runs successfully. Make sure that your program doesn't consume extra space and there shouldn't be any memory leakage or exceptions in your code.**

**Note:** You can use built-in string functionality like `strlen`, `strcpy` etc. where required. You do not need to re-write these functions.

#### **Partial Definition of class and Main Program:**

```
class MyList{
    //You cannot add or modify DATA members
    char** arr; //Array of Strings
    int size;    //No of Strings in List
};
void main()
{
    MyList list1; //Initially list will be empty.
    cout<<"list1 = "<<list1;    //Displays all the strings in list

    "Apple" + list1; //Adds new string "Apple" in list1
    cout<<"list1 = "<<list1;    //Displays all the strings in list

    "Banana" + list1; //Adds new string "Banana" in list1
    cout<<"list1 = "<<list1;

    "Peach" + list1; //Adds new string "Peach" in list1
    cout<<"list1 = "<<list1;

    MyList list2; //Initially list will be empty.
    cout<<"list2 = "<<list2;

    "I love Pakistan" + list2; //Adds new string "I love Pakistan" in list2
    cout<<"list2 = "<<list2;

    "Happy Programming" + list2; //Adds new string "Happy Programming" in list2
    cout<<"list2 = "<<list2;

    MyList list3 = list1+list2; //Creates another object of MyList having all the strings
of list 1 and 2
    cout<<"List 3 = "<<list3;
}
```

#### **Required Output:**

```
list1 = []
list1 = [Apple]
list1 = [Apple, Banana]
list1 = [Apple, Banana, Peach]
list2 = []
list2 = [I love Pakistan]
list2 = [I love Pakistan, Happy Programming]
List 3 = [Apple, Banana, Peach, I love Pakistan, Happy Programming]
```